How does SU(2) represent rotations?

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1 Introduction

Normally we consider n-dimensional rotations to be associated with $SO(n)$, but as we know, 2-dimensional rotations can also be characterised with unit length complex numbers. This implies the following isomorphism:

$$U(1) \cong SO(2) \tag{1}$$

Exponentiating the $\mathfrak{u}(1)$ basis vector i generates these rotations. The isomorphism to $\mathfrak{so}(2)$ yields a matrix representation of i, which can be exponentiated to produce a 2-dimensional rotation matrix.

2 Quaternions

The natural next step may be to ask whether an analogy can be drawn to quaternions.[1] Let us first probe how exactly we can use these to parametrise 3-dimensional rotations, consider first a quaternion $\mathbf{q}$ only of the coordinate $\mathbf{i}$, we will exponentiate this and use it to act on the $\mathbf{i}$ basis vector, logically, this should yield no change, after all, rotating an x-axis vector around the x-axis should change nothing.

$$\mathbf{q} = e^{\mathbf{i}\theta} = \cos\theta + \mathbf{i}\sin\theta \tag{2}$$

Applying the rotation yields:

$$\mathbf{qi} = (\cos\theta + \mathbf{i}\sin\theta)\mathbf{i} = \mathbf{i}\cos\theta + \mathbf{i}^2\sin\theta = \mathbf{i}\cos\theta - \sin\theta \tag{3}$$

Notice that we have run into a problem, there exists an extra dimension in this final expression, one that is not tied to $\mathbf{i}$. To amend this, we can instead apply a rotation of θ through two half-angle $\mathbf{q}$ rotations in the form:

$$\mathbf{i}' = (\mathbf{qi})\mathbf{q}^{-1} \tag{4}$$

[1] Quaternions are an example of hypercomplex numbers, an extension of complex numbers to higher dimensions.

Since $\mathbf{i}^{-1} = -\mathbf{i}$, we can find that:

$$(\mathbf{qi})\mathbf{q}^{-1} = (\mathbf{i}\cos\theta/2 - \sin\theta/2)(\cos\theta/2 - \mathbf{i}\sin\theta/2) = \mathbf{i}(\cos^2\theta/2 + \sin^2\theta/2) = \mathbf{i} \tag{5}$$

This yields exactly what we want. By exponentiating any one of the $\mathbf{i}, \mathbf{j}, \mathbf{k}$ basis vectors of the quaternions, we can specify any 3-dimensional rotation.

3 The symplectic group $Sp(1)$

Just as unitary complex numbers live in $U(1)$, unitary quaternions live in $Sp(1)$.[2] You may then expect the following isomorphism to hold:

$$Sp(1) \cong SO(3) \tag{7}$$

But in fact, this is not the case. We will see why a little bit later. For now though, we know that there is a valid isomorphism:

$$Sp(1) \cong SU(2) \tag{8}$$

Since $\mathbf{i}, \mathbf{j}, \mathbf{k}$ form a basis of the lie algebra $\mathfrak{sp}(1)$, we find that the basis vectors of $\mathfrak{su}(2)$ form a matrix representation of them. You may know already of these basis vectors, they are the Pauli matrices.

4 The 3-dimensional rotation group $SO(3)$

The structure of the Lie algebra $\mathfrak{so}(3)$ is found to be:

$$[J_i, J_j] = \varepsilon_{ijk} J_k \tag{9}$$

We find that the Lie algebra $\mathfrak{su}(2)$ is remarkably similar:

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k \tag{10}$$

Therefore in scaling σ_i by $(2i)^{-1}$, we can obtain a representation of SO(3) in terms of the Pauli matrices, and thus in terms of quaternions. We say the following isomorphism holds:

$$\mathfrak{so}(3) \cong \mathfrak{su}(2) \tag{11}$$

5 Rotations in $SU(2)$

We can now form an element of $SO(3)$ in terms of the scaled Pauli matrices, we can write this generally as:

$$R = \exp\left(\frac{\theta}{2i}\sigma \cdot \hat{n}\right) \tag{12}$$

[2] $Sp(1)$ is given as:

$$Sp(2n) = \{M_{2n\times 2n} : M^T J M = J\} \tag{6}$$

Where J is a skew symmetric bilinear form called the *symplectic* form.

Where $\sigma \cdot \hat{n}$ is given as:

$$\sigma \cdot \hat{n} = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z \tag{13}$$

And $\hat{n}$ is the normal vector in the direction of the axis by which the rotation will proceed around. So we can already see here that rotations under the $SU(2)$ representation only individually give half of the power one may expect, this being a hint towards the unusual rotation properties of spinors.[3] So, recalling the prior statement on how quaternion rotations must be performed (to preserve the structure of 3-dimensional space), provided a vector U, the full rotation under $SU(2)$ would be given as:

$$U \to U' = R^\dagger U R \tag{14}$$

6 Why $Sp(1) \ncong SO(3)$?

Since we know that there exists an isomorphism between $Sp(1)$ and $SU(2)$, if there also existed one between $SU(2)$ and $SO(3)$ then our initial statement would hold. The reason that this is not the case lies in the unusual rotation property of quaternions, and thus the Pauli matrices. (Specified in section 2 and eqn 14.) Suppose an element $A \in SO(3)$ representing a unique rotation. It acts on an object V such that:

$$V \to V' = AV \tag{15}$$

That same rotation represented in $SU(2)$ uses the element $B \in SU(2)$ such that:

$$V \to V' = B^\dagger V B \tag{16}$$

But notice that $-B$ also gives the same rotation:

$$V \to V' = (-B^\dagger)V(-B) = B^\dagger V B \tag{17}$$

Therefore, there is not an isomorphism between $SU(2)$ and $SO(3)$ because the map is not 1-to-1. There are *two* valid elements of $SU(2)$ for any one in $SO(3)$. For this reason, we call $SU(2)$ the *double cover* of $SO(3)$.

[3] Recall that spinors represent spin-1/2 particles, and the Pauli matrices act as the spin operators for the spin-1/2 representation.